

MATH 301: Homework 2

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Question 1: Section 2.1-10

Part (a): If $a < b$ and $c \leq d$, prove that $(a + c) < (b + d)$

Suppose $a < b$ and $c \leq d$. Consider the following:

Recall: (*Theorem*) *Rules of Inequality*

Let $a, b, c \in \mathbb{R}$. The following statements must hold:

- $(a > b) \wedge (b > c) \implies a > c$
- $(a > b) \wedge (c > 0) \implies ca > cb$
- $(a > b) \implies (a + c) > (a + b)$
- $(a > b) \wedge (c < 0) \implies ca < cb$

Applying the theorem above, consider the following:

$$\begin{aligned}
 a &< b \\
 a + c &< b + c \\
 b + c &\leq b + d \\
 a + c &< b + c \leq b + d \\
 a + c &< b + d
 \end{aligned}$$

Part (b): If $0 < a < b$ and $0 \leq c \leq d$, prove that $0 \leq ac \leq bd$

Suppose $0 < a < b$ and $0 \leq c \leq d$

Recall: (*Theorem*) *Rules of Inequality*

Let $a, b, c \in \mathbb{R}$. The following statements must hold:

- $(a > b) \wedge (b > c) \implies a > c$
- $(a > b) \wedge (c > 0) \implies ca > cb$
- $(a > b) \implies (a + c) > (a + b)$
- $(a > b) \wedge (c < 0) \implies ca < cb$

Ordering If $a - b \in P$, we write $a > b$ (or $b < a$).

If $a - b \in P \cup \{0\}$, we write $a \geq b$ (or $b \leq a$).

For any two real numbers a, b exactly one of the following holds:

$$a > b, a < b, a = b$$

Recall: *Properties of Positive Sets*

Let $P \neq \emptyset, P \subseteq \mathbb{R}$ be called the set of positive real numbers.

1. If $a, b \in P$ then $a + b \in P$.
2. If $a, b \in P$ then $a \cdot b \in P$.
3. *Trichotomy Property:* If $a \in \mathbb{R}$, then exactly one of the following must be true:

$$a \in P, \quad a = 0, \quad -a \in P$$

Applying the theorems above, observe the following:

$$(0 < a) \wedge (0 \leq c) \implies (0 \leq ac)$$

Thus the following statements hold:

$$\begin{aligned} 0 &\leq c \leq d \\ 0 &\leq bc \leq bd \\ 0 &< a < b \\ 0 &\leq ac \leq bc \leq db \\ 0 &\leq ac \leq db \end{aligned}$$

Question 2: Section 2.1-11

Part (a): If $(a > 0)$, then $(1/a > 0)$ and $\frac{1}{1/a} = a$

Suppose $a > 0$ and $\frac{1}{a} \not> 0$. Consider the following:

$$\begin{aligned} \frac{1}{a} &\not> 0 \\ \left(\frac{1}{a}\right) \cdot a &\not> 0 \cdot a \\ \left(\frac{1}{a}\right) \cdot a &\not> 0 \\ 1 &\not> 0 \end{aligned}$$

Clearly $1 > 0$ thus this is a contradiction.

Thus we must accept the original proposition: $a > 0 \implies \frac{1}{a} > 0$.

Prove $\frac{1}{1/a} = a$:

$$\frac{1}{a} \cdot a = 1 \quad \text{Multiplicative Inverse}$$

$$\frac{1}{a} \cdot \frac{1}{\frac{1}{a}} = 1 \quad \text{Multiplicative Inverse}$$

$$a \cdot \frac{1}{a} \cdot \frac{1}{\frac{1}{a}} = 1 \cdot a \quad \text{Multiply } a \text{ on both sides}$$

$$1 \cdot \frac{1}{\frac{1}{a}} = 1 \cdot a \quad \text{Multiplicative Inverse}$$

$$\frac{1}{\frac{1}{a}} = a \quad \text{Multiplicative Identity}$$

Thus we can say $\frac{1}{1/a} = a$

Now we have proven if $(a > 0)$, then $(1/a > 0)$ and $\frac{1}{1/a} = a$.

Part (b): If $(a < b)$, then $a < \frac{1}{2}(a + b) < b$

Consider the following:

$$a = a$$

$$\frac{1}{2}a = \frac{1}{2}a$$

$$\frac{1}{2}a + \frac{1}{2}a = \frac{1}{2}a + \frac{1}{2}a$$

$$\frac{1}{2}(a + a) = \frac{1}{2}(a + a)$$

$$\frac{1}{2}(2a) = \frac{1}{2}(a + a)$$

$$\frac{1}{2}2(a) = \frac{1}{2}(a + a)$$

$$a = \frac{1}{2}(a + a)$$

$$a < b \implies a + a < a + b$$

$$a < b \implies a + b < b + b$$

$$a < \frac{1}{2}(a + b) < b$$

Question 3: Sec 2.1-13

If $a, b \in \mathbb{R}$, show that $a^2 + b^2 = 0$ iff $a = 0$ and $b = 0$.

Case 1: $\rightarrow$

Prove: If $a, b \in \mathbb{R}$, show that $a^2 + b^2 = 0$ implies $a = 0$ and $b = 0$.

Consider the contrapositive statement, $a \neq 0$ or $b \neq 0$ implies $a^2 + b^2 \neq 0$:

Observe the following:

If $a \in \mathbb{P}$

$$a \in \mathbb{P} \implies a^2 = a \cdot a \in \mathbb{P} \implies a^2 > 0$$

If $-a \in \mathbb{P}$: (Consider $\forall n \in \mathbb{R}, (n - n) \cdot (-n) = 0$)

$$\begin{aligned}
n \cdot (-n) + (-n)(-n) &= 0 \\
(-n) + (-n)(-n) &= 0 \\
\implies (-n)(-n) &= n^2 \in \mathbb{P}
\end{aligned}$$

Therefore, $-a^2 \in \mathbb{P}$. The same logic can be applied to b .

$$\text{Observe If } a^2, b^2 \in \mathbb{P} \implies a^2 + b^2 \in \mathbb{P}.$$

Therefore we have proven the contrapositive statement if $a \neq 0$ or $b \neq 0$ implies $a^2 + b^2 \neq 0$

Case 2: $\leftarrow$

Prove: If $a, b \in \mathbb{R}$, show that $a = 0$ and $b = 0$ implies $a^2 + b^2 = 0$.

Consider the following:

$$0 \cdot 0 = 0 \implies \text{If } x = 0, \text{ then } x^2 = x \cdot x = 0 \cdot 0 = 0$$

As $a = b = 0$ then naturally if $a = 0$ and $b = 0$ implies $a^2 + b^2 = 0$

Question 4: Section 2.1-14

If $0 \leq a < b$, show that $a^2 \leq ab < b^2$. Show by example that it does **not** follow that $a^2 < ab < b^2$. Let $a = 0, b = 1$. Thus $0 \leq 0 < 1$ satisfy $a^2 \leq ab < b^2$. Observe $a^2 < ab < b^2$ does not hold as $0 < 0 < 1$ is false as $0 \not< 0$

Question 5: Section 2.1-20

Part a: If $0 < c < 1$, show that $0 < c^2 < c < 1$.

We can define $c < 1$ as $c = 1 - \epsilon$, where $\epsilon = \frac{1}{n}$ for some $n \in \mathbb{N}$ such that $0 < \epsilon < 1$.

Observe the following:

$$\begin{aligned}
1 &> 1 - \epsilon \\
c &= 1 - \frac{1}{n} \\
c^2 &= \left(1 - \frac{1}{n}\right)^2 = 1 - \frac{2}{n} + \frac{1}{n^2}
\end{aligned}$$

We want to compare c and c^2 . Since $n \geq 1$ and $\epsilon > 0$, multiplying the inequality $c < 1$ by c (where $c > 0$) preserves the sign:

$$\begin{aligned}
c \cdot c &< 1 \cdot c \\
c^2 &< c
\end{aligned}$$

Thus, combining with our initial range:

$$0 < c^2 < c < 1$$

Part b: If $1 < c$, show that $1 < c < c^2$.

We can define $c > 1$ as $c = 1 + \epsilon$, where $\epsilon = \frac{1}{n}$ for some $n \in \mathbb{N}$.

Observe the following:

$$\begin{aligned} 1 &< 1 + \epsilon \\ c &= 1 + \epsilon \\ c^2 &= (1 + \epsilon)^2 \\ &= 1 + 2\epsilon + \epsilon^2 \end{aligned}$$

Since $\epsilon > 0$, it must be that $\epsilon < 2\epsilon + \epsilon^2$. Therefore:

$$\begin{aligned} 1 &< 1 + \epsilon < 1 + 2\epsilon + \epsilon^2 \\ 1 &< c < c^2 \end{aligned}$$

Since we defined $\epsilon > 0$, the statement $1 < c < c^2$ holds.

Question 6: Section 2.1-23

If $a > 0$, $b > 0$, and $n \in \mathbb{N}$, show that $a < b$ iff $a^n < b^n$ (induct.)

$\forall a, b \in \mathbb{R}, a > 0, b > 0$ and $n \in \mathbb{N}$.

Consider $P(n) : a^n < b^n \iff a < b$

Observe $P(1)$:

$$\begin{aligned} a^1 &< b^1 \implies a < b \\ a &< b \implies a^1 < b^1 \end{aligned}$$

Suppose $P(k), k \in \mathbb{N}$:

$$a^k < b^k \implies a < b$$

Observe the following:

$$\begin{aligned} a^k &< b^k \\ a \cdot a^k &< b \cdot a^k & \text{Eq 1} \\ b \cdot a^k &< b \cdot b^k & \text{Eq 2} \end{aligned}$$

From Eq (1,2), we can derive the following

$$\begin{array}{r} a^{k+1} < b \cdot a^k \\ + \quad b \cdot a^k < b^{k+1} \\ \hline a^{k+1} + (ba^k) < (ba^k) + b^{k+1} \end{array}$$

Thus we can apply algebra to derive the final statement.

$$\begin{aligned} a^{k+1} + (ba^k) + (-ba^k) &< (ba^k) + (-ba^k) + b^{k+1} \\ a^{k+1} &< b^{k+1} \end{aligned}$$

Thus we conclude $a^k < b^k \implies a^{k+1} < b^{k+1}$. Therefore $a < b$ iff $a^n < b^n$.

Question 7: Section 2.1-25

Assuming the existence of roots, show that $c > 1 \implies c^{1/m} < c^{1/n}$ iff $m > n$

Case 1: $(c > 1 \implies c^{1/m} < c^{1/n}) \implies m > n$

Suppose $c > 1 \implies c^{1/m} < c^{1/n}$ and $m \not> n$.